

HR Analytics Dashboard

End-to-End Workforce Retention Analytics Solution

Project Overview

Employee attrition is a major workforce challenge that impacts productivity, recruitment costs, organizational knowledge, and business continuity. This project develops an end-to-end HR Analytics solution using the IBM HR Employee Attrition Dataset to identify workforce risks, uncover attrition drivers, discover high-risk employee personas, and provide data-driven retention recommendations.

Business Problem

HR leaders need answers to critical workforce questions:

- Which employee groups are most likely to leave?
- Which workforce segments contribute most to attrition?
- What factors influence employee turnover?
- Which employee personas require retention intervention?
- Where should retention investments be prioritized?

Traditional reporting answers “what happened” but rarely explains “why it happened.”

Business Objectives

The project aims to:

- Measure workforce health.
- Analyze employee attrition patterns.
- Identify key attrition drivers.
- Discover high-risk employee personas.
- Prioritize retention initiatives.
- Support workforce planning decisions.

Dataset Overview

Dataset: IBM HR Employee Attrition Dataset

Records: 1,470 Employees

Target Variable:

- Attrition (Yes / No)

Data Domains:

- Demographics
- Job Information
- Compensation
- Employee Experience
- Career Progression
- Workforce Retention

Project Methodology

Data Assessment ↓ Data Cleaning ↓ Feature Engineering ↓ Exploratory Data Analysis ↓ Driver Analysis ↓ Contribution Analysis ↓ Risk Matrix ↓ 2D Analysis ↓ 3D Persona Analysis ↓ Recommendations ↓ Dashboard Design

Feature Engineering

The following analytical fields were created:

- Age Band
- Income Band
- Tenure Band
- Promotion Gap

These derived features improved workforce segmentation and enabled deeper retention analysis.

Key KPIs

Workforce KPIs

- Total Employees
- Average Age
- Average Monthly Income
- Average Years at Company

Attrition KPIs

- Attrition Rate
- Exit Count
- Segment Attrition Rate

Advanced Analytics KPIs

- Attrition Index
- Contribution Percentage
- Highest Risk Segment
- Largest Contributor
- High-Risk Persona

Analytical Framework

Driver Analysis

Purpose:

Identify workforce segments with the highest attrition risk.

Metrics:

- Employees
- Exits
- Attrition Rate
- Attrition Difference
- Attrition Index

Contribution Analysis

Purpose:

Identify workforce segments responsible for the largest share of exits.

Metric:

Contribution % = Segment Exits ÷ Total Exits

Risk Matrix

Purpose:

Combine attrition risk and workforce impact.

Outputs:

- Highest Risk Segment
- Largest Contributor
- Exit Volume
- Attrition Rate

Two-Dimensional Analysis

Purpose:

Identify combinations of workforce drivers associated with elevated attrition.

Example:

OverTime + Low Income

58.49% Attrition

Three-Dimensional Persona Analysis

Purpose:

Identify employee personas exhibiting the highest attrition risk.

Example:

Low Income

- OverTime
- New Hire

81.25% Attrition

Key Findings

Attrition Drivers

Highest Risk Segments:

Driver	Highest Risk Segment	Attrition Rate
Job Role	Sales Representative	39.76%
Tenure	New Hire	34.88%
Age	18-25	34.78%
Work-Life Balance	Level 1	31.25%
OverTime	Yes	30.53%

Largest Contributors

Driver	Largest Contributor	Exits
Department	Research & Development	133
Job Role	Laboratory Technician	62
Income Band	Low Income	108
Job Level	Level 1	143

High-Risk Personas

Persona	Attrition Rate
Low Income + OverTime + New Hire	81.25%
Laboratory Technician + OverTime + New Hire	75.00%
Sales + Job Level 1 + OverTime	72.73%

Business Insights

- Attrition risk is highest among early-career employees.
- Overtime significantly increases turnover risk.
- Low-income employees exhibit elevated attrition.
- Entry-level employees experience the greatest workforce loss.
- Sales Representatives represent the highest-risk job role.
- Workforce risk is influenced by multiple interacting factors rather than a single driver.

Recommendations

Immediate Actions

Improve New Hire Retention

- Structured onboarding
- Early-career mentoring
- First-year engagement programs

Review Overtime Practices

- Monitor workload distribution
- Improve workforce planning
- Reduce burnout risk

Evaluate Compensation

- Review low-income employee segments
- Assess internal pay competitiveness

Strategic Actions

Career Development Programs

Focus on:

- Entry-level employees
- Young workforce segments

Manager Enablement

Provide retention coaching for managers supervising high-risk groups.

Data Limitations

- Dataset represents a single workforce snapshot.
- Historical attrition trends are unavailable.
- Exit reasons are not available.
- Financial cost data is unavailable.
- Findings support descriptive and diagnostic analytics only.

Deliverables

Analytical Assets

- Data Assessment
- Data Cleaning Report
- EDA Notebook
- Driver Analysis
- Contribution Analysis
- Risk Matrix
- Persona Analysis

Documentation

- BRD
- FRD
- User Stories
- KPI Dictionary
- Insight Log
- Recommendation Log

Dashboard

Excel-based HR Analytics Dashboard

Project Outcome

The project successfully transformed employee data into actionable retention intelligence by identifying workforce risks, discovering attrition drivers, prioritizing retention opportunities, and providing a structured framework for workforce decision-making.